

Week 1 Writing Assignment (CS-EE5841)

Data Preprocessing and Perceptron

Total Points: 42

Instructions: Complete all **Tasks** for each question (Q1-Q3). Show all key steps using formulas and provide brief interpretation where needed.

Q1. Z-Score Outlier Detection (15 pts)

Question

Given dataset:

$$X = \{5, 3, 7, 8, 100\}$$

Mean: $\mu = 5.6$, Standard deviation: $\sigma = 3.0$

Tasks

- Compute the Z-score for $X = 100$
- Determine whether 100 is an outlier

Hint

$$Z = \frac{X - \mu}{\sigma}$$

Solution

Step 1: Compute the Z-score

Using the given mean $\mu = 5.6$ and standard deviation $\sigma = 3.0$, we substitute $X = 100$ into the Z-score formula:

$$\begin{aligned} Z &= \frac{100 - 5.6}{3.0} \\ Z &= \frac{94.4}{3.0} \\ Z &\approx 31.47 \end{aligned}$$

Step 2: Determine if it is an outlier

A standard rule of thumb in statistics is that any data point with a Z-score greater than 3 or less than -3 is considered an outlier. Since $31.47 \gg 3$, the value 100 is definitively an outlier.

Interpretation: The Z-score measures how many standard deviations a data point is away from the mean. A Z-score of approximately 31.47 indicates that the value 100 is extremely far from the rest of the dataset's distribution, confirming it is an anomaly.

Q2. Min-Max Normalization (12 pts)

Question

Given dataset:

$$X = [10, 15, 20, 25, 30]$$

Tasks

- Apply Min-Max normalization to the dataset
- Compute normalized value for $X = 20$

Hint

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Solution

Step 1: Apply Min-Max normalization

First, identify the minimum and maximum values in the dataset:

$$X_{min} = 10, \quad X_{max} = 30$$

The range (denominator) is:

$$X_{max} - X_{min} = 30 - 10 = 20$$

Now, apply the formula to each element in X :

$$X_{1,norm} = \frac{10 - 10}{20} = 0$$

$$X_{2,norm} = \frac{15 - 10}{20} = 0.25$$

$$X_{3,norm} = \frac{20 - 10}{20} = 0.5$$

$$X_{4,norm} = \frac{25 - 10}{20} = 0.75$$

$$X_{5,norm} = \frac{30 - 10}{20} = 1$$

The fully normalized dataset is: $[0, 0.25, 0.5, 0.75, 1]$.

Step 2: Compute normalized value for $X = 20$

As calculated above, the normalized value for $X = 20$ is 0.5.

Interpretation: Min-Max normalization scales all data points linearly to fit within a specific range, typically $[0, 1]$. Because the original values are evenly spaced, their normalized counterparts are also evenly distributed between 0 (the minimum) and 1 (the maximum).

Q3. Perceptron Weight Update (15 pts)

Question

Given:

$$x = [2, 1], \quad y = 1$$

Initial weights:

$$w = [0.5, -0.5], \quad b = 0$$

Learning rate:

$$\eta = 0.1$$

Activation:

$$\hat{y} = \text{step}(w \cdot x + b)$$

Tasks

- Compute predicted output $\hat{y}$
- Compute error
- Update weights and bias

Hint

$$\text{Error} = y - \hat{y}$$

$$w_{\text{new}} = w + \eta \cdot \text{error} \cdot x$$

$$b_{\text{new}} = b + \eta \cdot \text{error}$$

Solution

Step 1: Compute predicted output $\hat{y}$

First, calculate the weighted sum $z = w \cdot x + b$:

$$z = (w_1 \cdot x_1) + (w_2 \cdot x_2) + b$$

$$z = (0.5 \cdot 2) + (-0.5 \cdot 1) + 0$$

$$z = 1 - 0.5 = 0.5$$

Next, apply the step activation function. Assuming the standard definition where $\text{step}(z) = 1$ if $z \geq 0$, and 0 otherwise:

$$\hat{y} = \text{step}(0.5) = 1$$

Step 2: Compute error

Using the given hint:

$$\text{Error} = y - \hat{y}$$

$$\text{Error} = 1 - 1 = 0$$

Step 3: Update weights and bias

Since the error is 0, the weight and bias update equations will result in no change:

$$w_{new} = [0.5, -0.5] + 0.1 \cdot 0 \cdot [2, 1]$$

$$w_{new} = [0.5, -0.5] + [0, 0] = [0.5, -0.5]$$

$$b_{new} = 0 + 0.1 \cdot 0$$

$$b_{new} = 0$$

Interpretation: The perceptron correctly classified the input x right out of the gate, outputting a 1 which perfectly matched the target label $y = 1$. Because there was no error in the prediction, the learning algorithm did not apply any updates to the weights or the bias.